NBA Shot Quality Capstone

Data Visualizations

DAT 490 — Data Science Capstone (2026 Summer A)

Team Members

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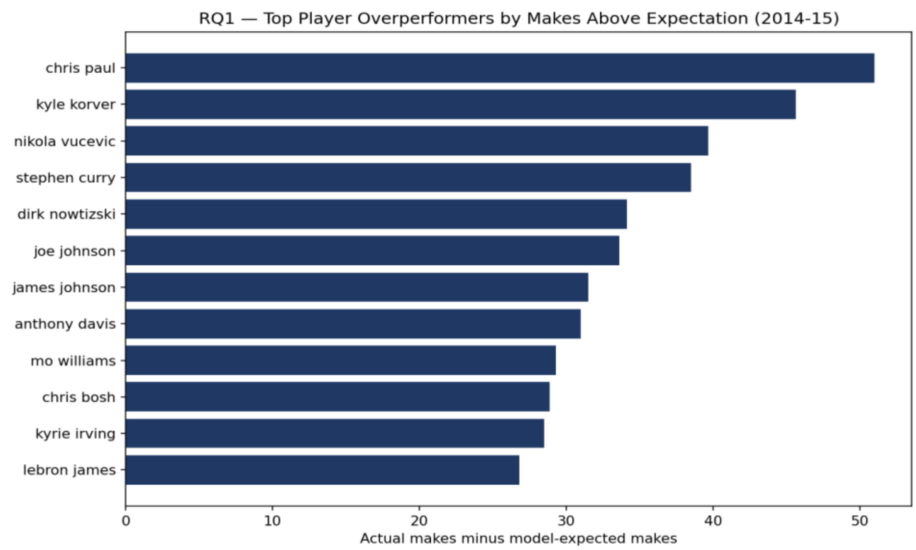
Data Visualizations

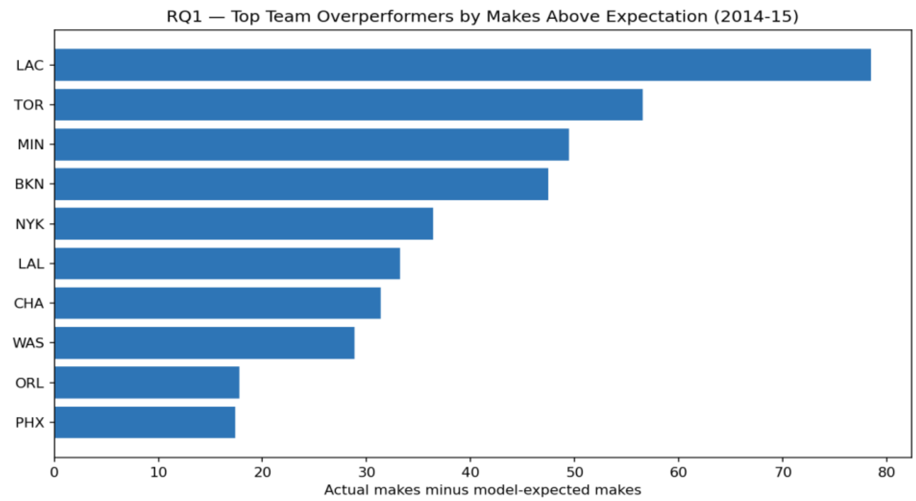
The visualizations in this section are built from the output of our modeling work, and they are meant to support the conclusions we draw for each research question rather than to explore the raw data the way our EDA did. Each chart is paired with a short explanation of what it shows and what it tells us. The figures are organized by research question, followed by a comparison of the two models we trained.

RQ1 — Which players and teams consistently outperform their expected shot value?

Our expected-shot-value model assigns every shot a probability of going in based on its context, which lets us compare how many shots a player actually made to how many the model expected them to make. The difference, summed across a player's season, is their makes above expectation.



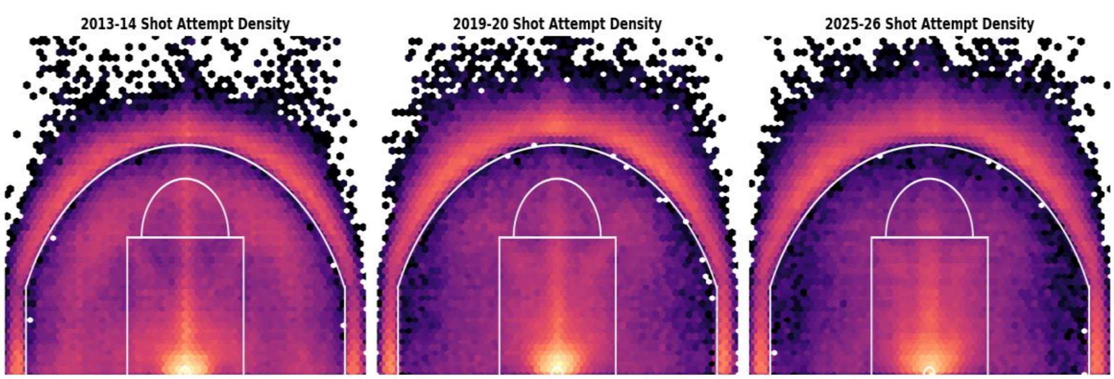
The top player overperformers in 2014-15 are a mix of elite shooters and skilled interior scorers. Chris Paul leads the group at roughly 51 makes above what the model expected, followed by Kyle Korver and Nikola Vučević. What stands out is that several of these players combine high volume with a meaningful per-shot edge — Korver, for example, made his shots at a rate about 9.5 percentage points higher than the model expected, which is an enormous gap for a player taking nearly 500 shots. These are the players the model identifies as beating the difficulty of the shots they take, rather than just taking easy shots.



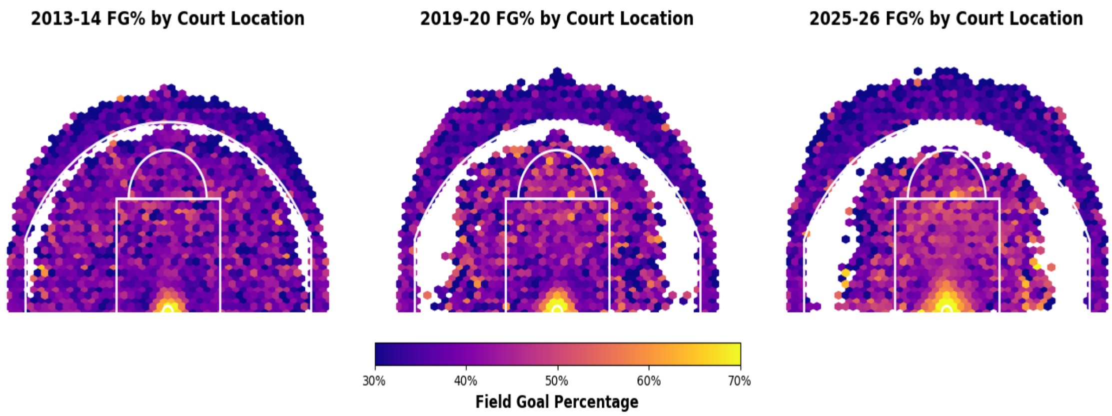
The same logic applies at the team level. The Los Angeles Clippers were the clear top overperforming team in 2014-15 at roughly 78 makes above expectation, well ahead of Toronto and Minnesota. This lines up with what we know about that Clippers roster, which paired elite shot creation with efficient finishing. Aggregating residuals to the team level smooths out the noise from any single player and shows which teams, as a whole, converted their shots better than the model predicted.

RQ2 — How has shot success by court location changed across the tracking era?

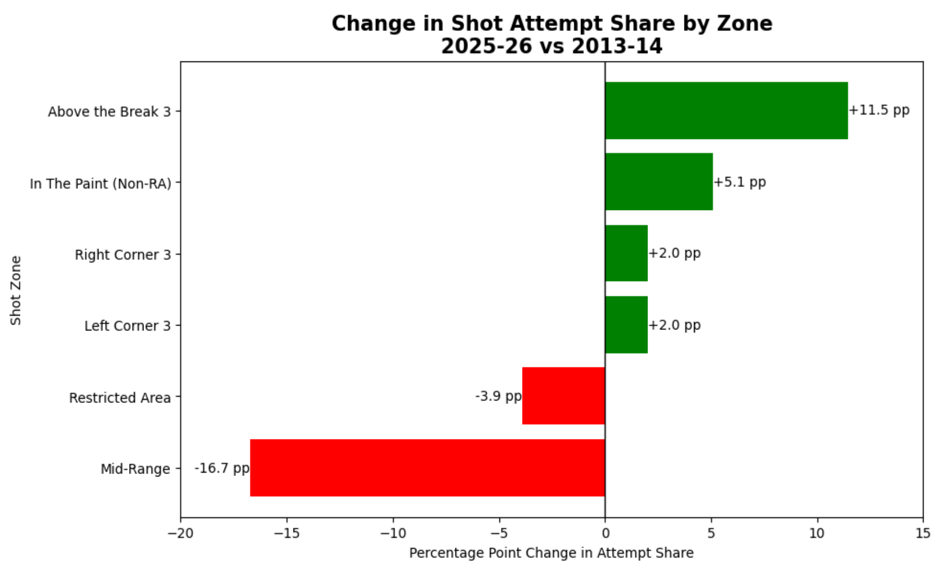
These visualizations trace the geography of shooting across the tracking era. They use the full thirteen seasons of location data rather than the single-season modeling sample, because the question is designed this way.



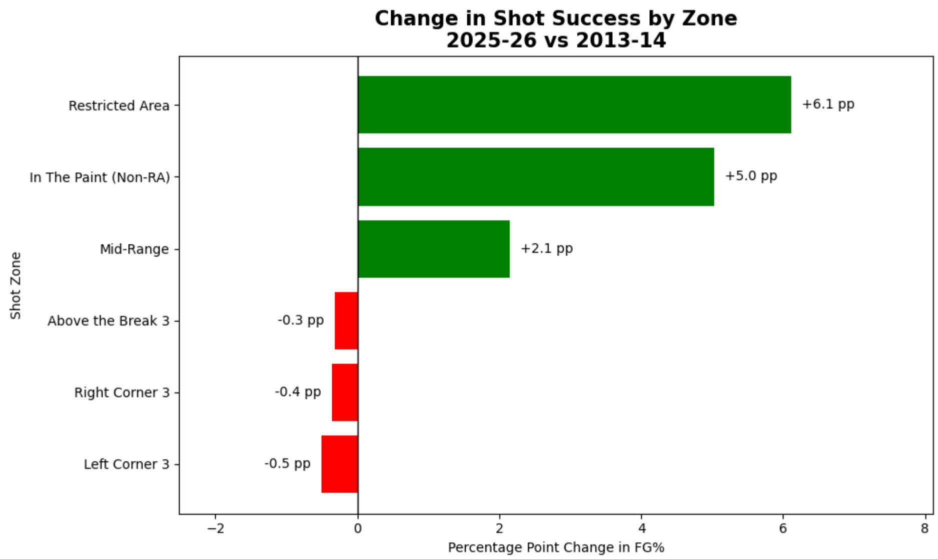
The shot-density maps make the three-point revolution impossible to miss. Moving from 2013-14 to 2019-20 to 2025-26, the dense cluster of attempts steadily migrates outward: the area just inside the arc thins out dramatically, while the volume beyond the three-point line and right at the rim intensifies. The mid-range, which is heavily populated in the 2013-14 panel, is visibly hollowed out by 2025-26.



Coloring the same court by field goal percentage instead of volume tells a complementary story. The restricted area stays consistently the hottest zone across all three seasons, and the corners remain reliably efficient three-point real estate. What is striking is how stable the efficiency surface is even as the volume map changes so much underneath it — teams moved their shots toward the most efficient zones rather than getting dramatically better at any one spot.



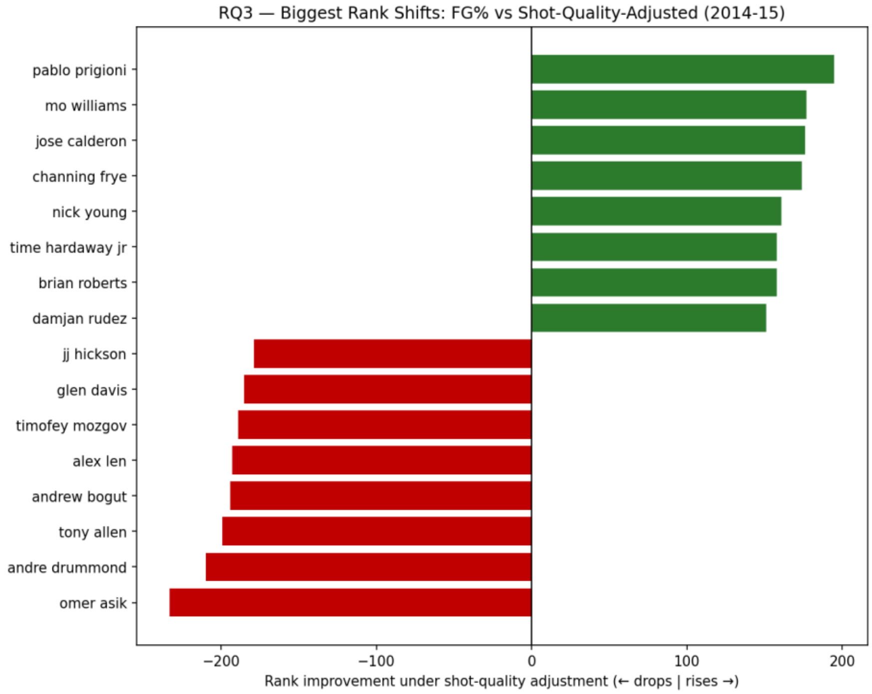
Quantifying the migration by zone confirms the visual story. Between 2013-14 and 2025-26, mid-range attempts fell by roughly 17 percentage points of total shot share — by far the largest move on the chart — while above-the-break threes rose by about 11 percentage points, and the corners and the area inside the arc picked up the rest. This is the clearest single summary of how shot selection has been rebuilt over the era.



Field goal percentage by zone changed far less than attempt share did, which reinforces the point that the era's shift was about where teams shoot rather than how well. The one notable exception is the restricted area, where shooting improved by about six percentage points — players are finishing at the rim more effectively than they did a decade ago, even as the rest of the efficiency landscape held roughly steady.

RQ3 — Does shot-quality-adjusted ranking differ from traditional efficiency metrics?

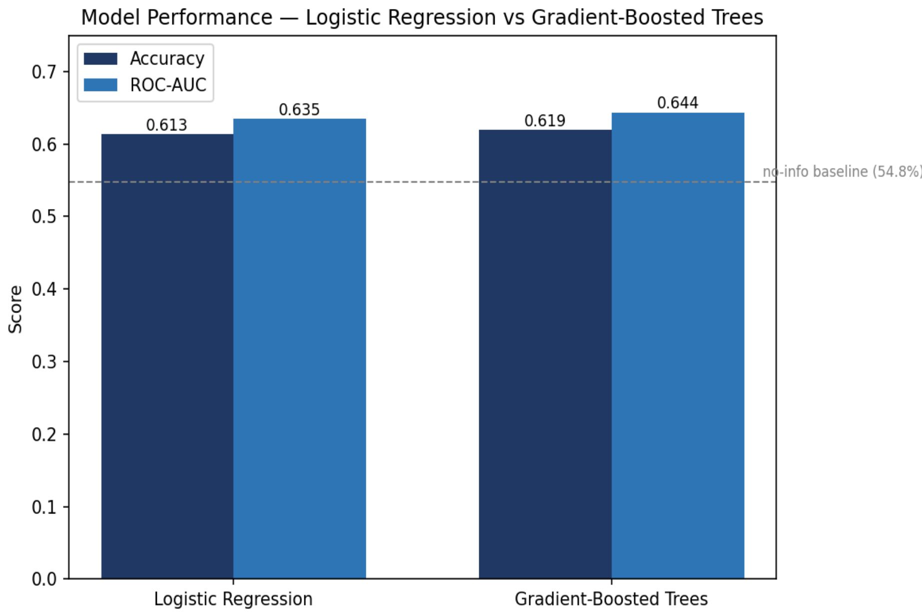
To answer this we ranked players two ways — by traditional field goal percentage, and by our shot-quality-adjusted metric — and measured how far each player moved between the two rankings.



The chart shows the players whose rankings change the most, and the pattern is clean. The biggest risers, shown in green, are perimeter shooters: Pablo Prigioni climbs about 195 spots, with Mo Williams, Jose Calderon, and Channing Frye close behind. These are players whose raw field goal percentage looks unremarkable but who consistently make difficult, contested perimeter shots the model did not expect to go in. The biggest fallers, shown in red, are interior players: Omer Asik drops about 233 spots, with Andre Drummond and others nearby. Their raw field goal percentages are high because they take easy shots near the basket, but once the model accounts for how makeable those shots were, the edge disappears. The fact that the two ends of the chart sort so cleanly by player type is the strongest evidence that the shot-quality-adjusted ranking tells a much different story than traditional efficiency metrics do.

Model performance

Underlying all of the player and team results is the shot-success model itself, so it is worth showing that the model works.



We trained two models on the same data: a logistic regression baseline and a gradient-boosted tree model. Both clear the no-information baseline of 54.8 percent — the accuracy you would get by always guessing the more common outcome — which confirms that shot context carries good predictive signal. The gradient-boosted model is the stronger of the two on both measures, reaching about 61.9 percent accuracy and a ROC-AUC of about 0.644, compared to 61.3 percent and 0.635 for the logistic baseline. The improvement is modest but consistent, which is why the boosted model is the one we use to generate the expected-make probabilities behind every other chart in this section.